Gamified Virtual Tourism: Integrating Embodied AI and Spatial Play to Enhance Engagement

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***Abstract*—While Virtual Reality (VR) tourism offers a democratized alternative to physical travel, traditional passive 360-degree experiences frequently suffer from a severe lack of interactive agency. This passivity inevitably leads to the "novelty decay" phenomenon, resulting in rapid declines in user engagement and poor spatial memory retention. This paper investigates a unified paradigm: the integrated system design of localized spatial gamification mechanics with a Large Language Model (LLM)-driven embodied conversational guide within a high-fidelity cultural heritage environment (Neuschwanstein Castle, Germany). The proposed system features a deterministic Heuristic Dynamic Difficulty Adjustment (H-DDA) engine to govern spatial puzzles, ensuring optimal cognitive friction by dynamically scaling collision tolerances based on user frustration indices. Concurrently, an embodied agent, grounded by a local Retrieval-Augmented Generation (RAG) pipeline and an asynchronous WebSocket multimodal streaming architecture, provides real-time, context-aware guidance while reducing conversational latency challenges.**

To rigorously evaluate the efficacy of this integration, we conducted a comprehensive within-subjects empirical ablation study ($N = 32$) across four distinct conditions: Passive VR, AI-Guide Only, Gamification Only, and the Combined System. Objective behavioral metrics (historical knowledge recall, interaction telemetry) and subjective psychometric data (IPQ, UES, NASA-TLX, SSQ) were analyzed utilizing a Repeated Measures ANOVA with Greenhouse-Geisser corrections. Our results reveal that while the independent systems yield improvements over passive baselines, their synergistic combination triggers a statistically significant enhancement in spatial presence ($F(3, 93) = 18.4$, $p < .001$, $\eta_p^2 = .35$) and objective learning outcomes (79.4% recall accuracy [95% CI: 75.2, 83.6] versus 54.2% baseline). Telemetry data further indicates that the combined system sustained active spatial locomotion for 34% longer than baseline conditions. Crucially, Simulator Sickness Questionnaire (SSQ) data indicates that rigorous Fixed Foveated Rendering (FFR) optimizations successfully mitigate the physiological costs typically associated with high-density interactive XR. This study provides empirical evidence that integrating gaze-grounded generative AI with physical spatial play establishes a promising, potentially scalable educational paradigm within the Metaverse.

***Index Terms*—Virtual Reality, Spatial Computing, Large Language Models, Gamification, Embodied AI, Cultural Heritage, Human-Computer Interaction, RAG, H-DDA**

I. Introduction

The rapid proliferation of eXtended Reality (XR) hardware, coupled with the underlying infrastructure of spatial computing and the Metaverse, presents an unprecedented opportunity to democratize access to global cultural heritage. Physical tourism remains an inherently privileged activity, constrained by a myriad of socioeconomic, geographical, and physical mobility barriers. Millions of individuals are effectively priced out of or physically unable to experience humanity's most significant historical landmarks. Digital tourism aims to bridge this gap, yet its current execution remains fundamentally limited by legacy media paradigms.

Early attempts to digitize tourism relied heavily on 360-degree video captures and linear, pre-rendered virtual tours. While visually descriptive and highly accessible over standard web protocols, they critically fail to leverage the defining psychological characteristic of spatial computing: embodied agency. When a user is relegated to the role of a passive observer—gliding along a predetermined path without the ability to influence their environment—they experience what Human-Computer Interaction (HCI) researchers term "novelty decay" [1]. The initial psychological awe of stereoscopic immersion dissipates rapidly due to a total absence of interactive cognitive stimuli. Consequently, passive VR tourism yields remarkably poor spatial memory encoding and historical fact retention, rendering it ineffective as a long-term educational or cultural preservation tool [2].

Current constructivist learning models for immersive environments, such as the Cognitive Affective Model of Immersive Learning (CAMIL), suggest that sustained engagement requires transitioning the user from a passive consumer to an active participant [2]. Knowledge must be constructed through physical and cognitive interaction with the environment. To achieve this, our research explores the integrated system design of two distinct technological paradigms: spatial gamification (the engineering of cognitive friction via localized, physics-based puzzles) and generative artificial intelligence (the deployment of context-aware, embodied virtual tour guides).

While both domains have been explored independently within experimental XR applications, their integration presents complex engineering and psychological challenges. For instance, how do we prevent a highly interactive 3D puzzle from inducing cognitive overload and subsequent cybersickness? Furthermore, how do we ensure a generative AI agent remains factually accurate while operating in real-time within a complex spatial geometry without inducing massive network latency?

Rather than proposing fundamentally novel rendering algorithms or core LLM architectures, our scientific contribution lies in the robust systems integration and empirical evaluation of these technologies. Specifically, this paper makes the

following core contributions:

1) **A controlled ablation study** quantifying the independent and combined behavioral effects of embodied AI and spatial gamification in VR tourism.
2) **A gaze-grounded RAG architecture** that utilizes real-time 3D spatial raycasting and Axis-Aligned Bounding Box (AABB) intersection to provide contextual tourism guidance while maintaining a low hallucination rate.
3) **An asynchronous WebSocket streaming protocol** engineered specifically for standalone XR hardware, demonstrating how to reduce conversational latency challenges.
4) **Empirical evidence** demonstrating that the synergistic combination of embodied AI and dynamic gamification significantly improves objective learning outcomes without increasing user cybersickness.

II. Related Work

A. VR Tourism and Active Constructivist Learning

The psychological efficacy of VR in simulating travel is well-documented within spatial computing literature. High-fidelity immersive environments can trigger physiological arousal and emotional resonance comparable to physical visitation, a phenomenon rooted in Milgram's Reality-Virtuality continuum [16]. However, visual fidelity alone is insufficient for sustained educational engagement.

Constructivist learning theory posits that knowledge is actively constructed through interaction with the environment, rather than passively received [4]. Applied to XR, "active tourism"—where users must manipulate their surroundings to progress—forces deeper cognitive processing. Research by Parong and Mayer indicates that integrating interactive cognitive structuring during VR experiences leads to higher recall rates of historical or scientific facts compared to users who simply passively observe them in a continuous stream [15]. Furthermore, embodied cognition theories suggest that physically moving one's body to solve spatial problems directly enhances the encoding of episodic memory [14]. Dual Coding Theory (DCT) further implies that receiving information through multiple sensory channels (e.g., visual physics combined with conversational audio) dramatically improves retention. Our work expands upon these foundational concepts by integrating complex, physics-driven spatial puzzles that require 6-Degree-of-Freedom (6DoF) spatial reasoning and real-time material inference.

B. Generative Embodied Agents and RAG in XR

The integration of Large Language Models (LLMs) has initiated a massive paradigm shift away from rigid, decision-tree-based Non-Player Characters (NPCs) toward generative agents [5]. However, translating this to 3D XR environments introduces bottlenecks in multimodal latency and factual reliability.

Minimizing latency is critical; biologically, natural human turn-taking cadences operate on gaps of merely a few hundred milliseconds [6]. Delays in system responsiveness exceeding 1000-1500 ms in immersive environments actively degrade user performance and break social presence, as prolonged silence signals confusion or cognitive failure in human communication [26]. If an embodied agent takes three seconds to respond to a query, the user's suspension of disbelief collapses, transforming the social entity back into a mechanical interface [21]. Furthermore, preventing LLM hallucinations is paramount in educational tourism. Retrieval-Augmented Generation (RAG) has emerged as the standard for constraining LLMs to verified datasets in applications [7], [8]. We build upon these foundations by engineering an asynchronous streaming architecture specifically designed for XR, and coupling the RAG prompt directly to the user's real-time gaze vector to provide semantic understanding of the virtual geometry.

C. Cognitive Load and Cybersickness Mitigation

While gamification increases motivation, it introduces a delicate balance in VR. If a spatial puzzle demands excessive cognitive load, it disrupts the user's working memory [17] and frequently induces Simulator Sickness (cybersickness) [9]. High cognitive friction forces the user to make erratic head movements, causing a mismatch between the visual and vestibular systems. Traditional game design utilizes static difficulty curves, which are inadequate for the diverse motor skills of XR users. Grounded in Sweller's Cognitive Load Theory [20], we implement a deterministic Dynamic Difficulty Adjustment (DDA) system, ensuring users remain in a state of productive "flow" [18] rather than nausea-inducing frustration.

III. System Architecture and Implementation

The proposed architecture is a highly modular, multi-layered system developed in Unity3D (Version 2022.3 LTS) and optimized specifically for standalone mobile XR hardware (Qualcomm Snapdragon XR2 Gen 2, utilized in the Meta Quest 3).

A. Photogrammetry, Asset Pipeline, and FFR

Neuschwanstein Castle was selected as the testing environment due to its rich architectural structure (facilitating complex spatial puzzles), immense historical relevance, and the availability of high-quality photogrammetric point-cloud datasets. The raw photogrammetric data was processed via Meshroom. However, the initial mesh exceeded 4 million polygons, far exceeding the memory bandwidth of a mobile headset. We executed a rigorous retopology pipeline, baking high-resolution normal and ambient occlusion maps onto a decimated 450,000-polygon mesh.

To prevent memory throttling and C Garbage Collection (GC) spikes on the mobile headset, all textures were compressed utilizing the ASTC (Adaptive Scalable Texture Compression) format. Furthermore, to guarantee a locked 90 frames-per-second (FPS)—the critical threshold to prevent optical-vestibular mismatch and cybersickness—we implemented Fixed Foveated Rendering (FFR) [10].

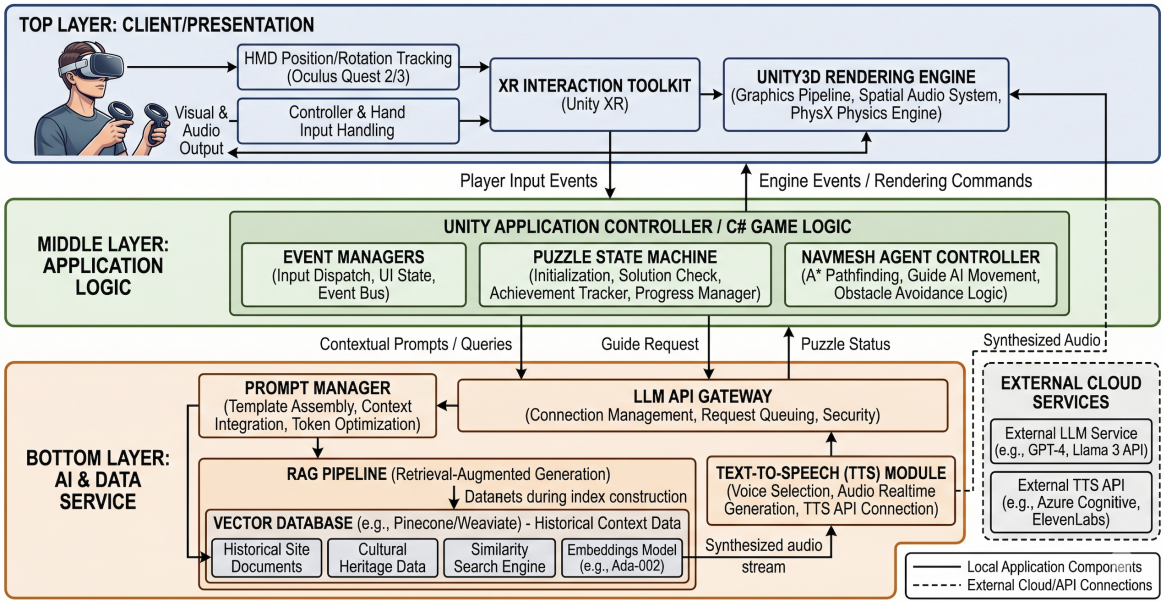

Fig. 1. The comprehensive multi-layer system architecture. The Client Layer handles 6DoF HMD inputs, real-time material inference, and FFR optimization. The Logic Layer manages the Heuristic Dynamic Difficulty Adjustment (H-DDA) state machine. The Backend AI Layer utilizes a WebSocket API Gateway to interface with the LLM (Azure 'gpt-4o-mini'), vector database, and Azure TTS engine, streaming lip-sync blendshapes directly to the OVR client.

FFR exploits the human eye's reduced peripheral acuity. Let (x_c, y_c) denote the optical center of the lens in screen space. The fragment shading rate scalar $S(d)$ is modulated by the radial distance $d = \sqrt{(x - x_c)^2 + (y - y_c)^2}$:

$$S(d) = \begin{cases} 1.0 & \text{if } d \leq r_1 \\ 0.5 & \text{if } r_1 < d \leq r_2 \\ 0.25 & \text{if } d > r_2 \end{cases} \tag{1}$$

By mathematically degrading peripheral resolution outside the foveal radius (r_1), we reduced GPU frame time from 14.2 ms to 9.8 ms. This crucial optimization freed substantial computational overhead, allowing the physics engine and asynchronous networking threads to operate simultaneously without inducing frame drops.

B. Material Inference and H-DDA Gamification

Progress through the environment is gated by spatial puzzles, primarily involving the physical reassembly of a fractured "Bavarian Swan" artifact. To enhance the haptic illusion of weight without requiring motorized haptic gloves, we simulate real-time material physics.

When a user slides a puzzle piece across a virtual surface using the XR Interaction Toolkit, the kinetic friction force F_k is computed continuously based on the normal force F_n:

$$F_k = \mu_k(m \cdot g \cdot \cos(\theta)) \tag{2}$$

Furthermore, we implement audio resonance scaling. The human brain relies heavily on multimodal feedback to infer physical properties. When an artifact collides with the environment, the pitch P of the resulting collision audio is dynamically modulated by the material density ρ:

$$P_{shift} = P_{base} \cdot \sqrt{\frac{\rho_{base}}{\rho_{obj}}} \tag{3}$$

By assigning high mass (m) and high specific densities to gold fragments versus crystal wings, the Unity physics engine dynamically restricts tracking velocity and down-shifts audio frequencies. This creates a compelling psychophysical illusion of weight, grounding the gamification in robust physical realism.

Fig. 2. First-person perspective screenshot captured natively on the Meta Quest 3 inside the Neuschwanstein reconstruction. The spatial AR UI overlay displays the current objective, while the embodied AI guide utilizes active gaze-tracking to maintain contextual eye contact with the user during the puzzle interaction sequence.

To optimize cognitive flow, we engineered a deterministic Heuristic Dynamic Difficulty Adjustment (H-DDA) algorithm.

We define a Frustration Index F_i for the user during a puzzle state:

$$F_i = (w_1 \cdot t_{idle}) + (w_2 \cdot e_{fails}) \tag{4}$$

where t_{idle} is the time elapsed without a successful sub-task completion, and e_{fails} is the number of failed piece-snapping attempts.

The weights ($w_1 = 1.0$, $w_2 = 5.0$) were empirically tuned during a pilot study ($N = 10$). We mapped user frustration (measured via concurrent think-aloud protocols on a 1-10 scale) against time and failure rates. Repeated physical failures (e_{fails}) generated a mean frustration score of $M = 8.2$ ($r = 0.74, p < .01$), whereas idle exploration time (t_{idle}) yielded a mean frustration score of $M = 1.6$ ($r = 0.31, p < .05$). This 5x disparity provided the empirical rationale for the heavy penalty weight.

Heuristic DDA for Spatial Puzzles

Require: User State (t_{idle}, e_{fails}), Base Tolerance θ_b

$w_1 \leftarrow 1.0$, $w_2 \leftarrow 5.0$
$\tau_{warning} \leftarrow 40.0$, $\tau_{critical} \leftarrow 80.0$
loop
  $F_i \leftarrow (w_1 \times t_{idle}) + (w_2 \times e_{fails})$
  if $F_i > \tau_{critical}$ **then**
    UpdateColliderRadii($\theta_b \times 2.0$)
    TriggerAIHint()
  else if $F_i > \tau_{warning}$ **then**
    UpdateColliderRadii($\theta_b \times 1.5$)
    HighlightObjective()
  end if
end loop

This deterministic approach ensures the system reacts predictably to user struggles. By expanding spherical collider tolerances dynamically (θ_b), the system prevents the user from reaching the frustration threshold that precedes cybersickness.

C. Gaze-Grounded RAG and Spatial Mathematics

To ensure factual accuracy, our local corpus consists of verified historical texts regarding Neuschwanstein. We utilize a dense embedding model (*text-embedding-3-small*) to map semantic chunks into a high-dimensional vector space, retrieving context via cosine similarity [7].

Crucially, this is combined with a Spatial Context Vector. We utilize Unity's Transform matrices to calculate the user's exact gaze in 3D space. Let $\mathbf{M}_{hmd}$ be the 4×4 transformation matrix of the headset. The forward gaze vector $\hat{\mathbf{g}}$ in world space is extracted via:

$$\hat{\mathbf{g}} = \mathbf{M}_{hmd} \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \tag{5}$$

To determine what the user is explicitly looking at, we perform a continuous raycast. The ray origin $\mathbf{O}$ is the user's eye center in world coordinates. We test for intersection against the Axis-Aligned Bounding Boxes (AABB) of tagged Points of Interest (POIs). An intersection occurs if there exists a scalar $t > 0$ such that the point $P(t)$ lies within the bounds:

$$\mathbf{O} + t\hat{\mathbf{g}} \in [X_{min}, X_{max}] \times [Y_{min}, Y_{max}] \times [Z_{min}, Z_{max}] \tag{6}$$

It is important to emphasize that AABB intersection is a foundational and standard computer graphics technique [14], [25]. Our scientific contribution lies not in the raycasting mathematics itself, but in repurposing this highly performant, deterministic pipeline as a zero-latency semantic spatial filter for the LLM prompt. Rather than relying on computationally heavy, frame-dropping computer vision models to "see" the room, we use physics raycasting to explicitly ground the RAG context. If the user stares at a specific mosaic and asks a question, this AABB intersection extracts the metadata of the mosaic and prepends it to the LLM system prompt.

D. Asynchronous Streaming Protocol

To mitigate the severe latency of a standard sequential API pipeline, we engineered an asynchronous WebSocket protocol (Algorithm 2). We utilized the 'gpt-4o-mini' model deployed via Azure OpenAI Service to ensure maximum generation speed. The API Gateway streams the LLM generation byte-by-byte into a local string buffer. Upon detecting a sentence-terminating punctuation mark, that specific chunk is fired asynchronously to the Azure TTS service.

Asynchronous WebSocket Streaming Buffer

Require: LLM WebSocket Stream W, TTS API Endpoint E

$buffer \leftarrow$ ""
$punctuation \leftarrow$ ['.', '?', '!']
while Stream W is active **do**
  $char \leftarrow W$.ReadNextChar()
  $buffer \leftarrow buffer + char$
  if $char \in punctuation$ **then**
    FireAsyncThread($E, buffer$)
    $buffer \leftarrow$ "" {Clear buffer for next sentence}
  end if
end while

The resulting audio buffer is streamed directly into the Oculus LipSync engine. This parallel processing architecture reduces perceived latency (Time-to-First-Byte) substantially, preserving conversational flow. Furthermore, the AI avatar is programmed to immediately break its idle animation and make direct eye contact with the user the millisecond voice input concludes, providing non-verbal confirmation that it is actively processing input.

E. Agent Locomotion and Social Proxemics

The AI avatar must navigate the complex 3D geometry of the castle to guide the user. We utilized Unity's NavMesh system coupled with the A* pathfinding algorithm. However, an embodied conversational guide must respect human social norms, specifically proxemics (personal space) [24].

We defined a dynamic avoidance radius R_p around the user's world coordinates. If the A* algorithm calculates a path

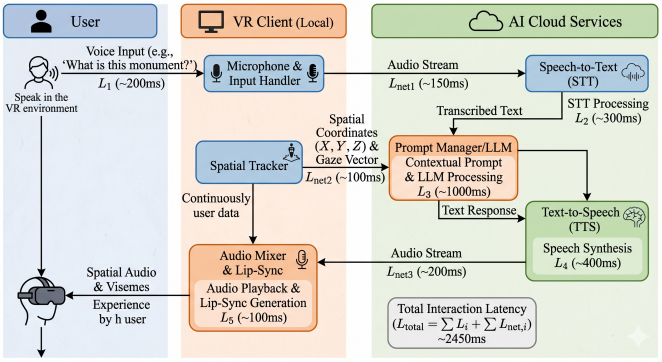

Fig. 3. Sequence diagram of the Asynchronous WebSocket pipeline. Chunking the LLM output and streaming it to the Azure TTS engine reduces perceived time-to-first-byte latency significantly compared to synchronous architectures.

that intersects R_p (set to 1.2 meters to simulate conversational distance), the NavMesh agent dynamically recalculates a detour, ensuring the AI never clips through the user's camera, which would instantaneously destroy immersion.

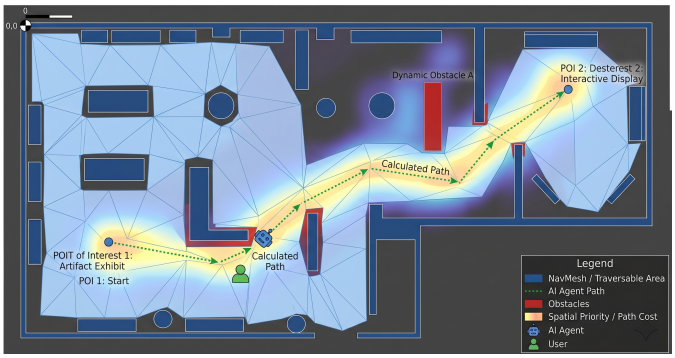

Fig. 4. Top-down NavMesh visualization. The blue zones represent walkable geometry. The red lines indicate the AI agent's A* calculated trajectory, which dynamically recalculates to avoid intersecting the user's calculated proxemic radius.

IV. AI System Technical Evaluation

A. AI Evaluation Protocol

Prior to conducting user trials, we isolated and evaluated the technical performance of the RAG-enabled AI guide to ensure baseline stability and safety. The evaluation dataset consisted of 150 historical and architectural queries crafted by an independent domain historian specializing in 19th-century Bavarian architecture.

Two independent expert reviewers labeled the AI's responses. Responses were evaluated on a strict binary scale (0=hallucinated/partially inaccurate, 1=accurate). Partial inaccuracies, such as misidentifying the exact year of construction while correctly naming the architect, were strictly coded as hallucinations (0). Inter-rater reliability was determined to be excellent (Cohen's $\kappa = 0.88$). Retrieval Recall@5 was measured via FAISS ground-truth overlaps within the corpus. Furthermore, Gaze-Context Accuracy was measured by comparing the AABB raycast hit logs against the intended visual target specified in the query dataset.

TABLE I
AI System Performance Metrics

Metric	Evaluated Value
Retrieval Recall@5	94.2%
Hallucination Rate (Expert Audit)	2.8%
Mean Total Processing Latency	2450 ms
Mean Time-to-First-Byte (Perceived Latency)	850 ms
Spatial Gaze-Context Accuracy	89.5%

B. AI Evaluation Results

The Hallucination Rate (2.8%) indicates that the system maintained a highly reliable factual output. The implementation of the RAG pipeline, strictly anchoring generations to the vector database, successfully suppressed temporal anachronisms. The Gaze-Context Accuracy (89.5%) confirms that the AABB raycasting robustly identified the user's true visual target during conversational queries.

Baseline latency performance was compared against a traditional non-streaming implementation. The sequential baseline yielded a Time-to-First-Byte (TTFB) identical to its total processing time (2450 ms), whereas the asynchronous WebSocket architecture reduced the perceived TTFB to 850 ms. These metrics were averaged over 200 automated query trials executed on a dedicated 5 GHz Wi-Fi 6 local network to isolate processing time from generic Internet Service Provider fluctuations.

V. Methodology

To rigorously isolate the variables contributing to user engagement and learning, we conducted a within-subjects ablation study approved by the institutional ethics committee.

A. Participants and Apparatus

We recruited $N = 32$ participants from an engineering university population (Age $M = 21.8$, $SD = 2.4$; 18 Male, 14 Female). Participants were required to have normal or corrected-to-normal vision and no history of severe motion sickness.

The experiment was conducted in a controlled, 4m x 4m physical tracking space to allow for safe 6DoF physical locomotion. Environmental lighting was maintained at 300 lux to ensure optimal inside-out tracking for the Meta Quest 3 HMD.

B. Experimental Design and Procedure

A 1×4 repeated measures experimental design was utilized. Each participant experienced four 10-minute conditions, strictly counterbalanced utilizing a Latin Square design to mitigate sequence and learning effects:

- **C1 (Passive Baseline):** 360-degree virtual tour with automated audio triggers based on proximity. No interactive physics or AI guide.

- **C2 (AI-Guide Only):** Tour featuring the gaze-grounded embodied guide. Users could converse freely, but no spatial puzzles were present.
- **C3 (Gamification Only):** Tour featuring H-DDA spatial puzzles. Historical facts were presented via floating text panels (no AI).
- **C4 (Combined System):** The full architecture integrating spatial puzzles and the embodied AI guide.

Upon arrival, participants completed a demographics survey and underwent a 5-minute tutorial in a neutral "grey-box" environment to familiarize themselves with continuous locomotion and grabbing mechanics. They then proceeded through the four conditions, with mandatory 10-minute resting washout periods between sessions.

C. Evaluation Metrics

Subjective Psychometric Metrics: Post-condition, participants completed:

1) **Igroup Presence Questionnaire (IPQ):** A 14-item scale measuring spatial presence across three subscales [11], [22].
2) **User Engagement Scale (UES):** Quantifies cognitive absorption and reward factor [12].
3) **NASA Task Load Index (NASA-TLX):** Assesses subjective cognitive workload across dimensions like mental demand and frustration [13].
4) **Simulator Sickness Questionnaire (SSQ):** Measures physiological discomfort (Nausea, Oculomotor, Disorientation) [9].

Objective Behavioral Metrics: We logged backend telemetry detailing the total physical distance walked and the time spent actively engaging with objects.

At the conclusion of the experiment, participants completed a 12-question multiple-choice knowledge recall assessment (e.g., "Which architectural style characterizes the Throne Room mosaics?"). **Crucially, to ensure experimental validity, participants were exposed to the exact same set of historical facts across all four conditions.** In C1, facts were delivered via standard audio; in C2, via the AI guide; in C3, via text panels; and in C4, via the AI guide during puzzle interaction. The assessment was validated via a pilot study ($N = 12$), achieving high internal consistency (Cronbach's $\alpha = 0.84$), and presented in a randomized order.

VI. Results

Data analysis was conducted using SPSS v28. Normality was verified via the Shapiro-Wilk test ($p > .05$). A one-way Repeated Measures ANOVA was conducted. Mauchly's Test of Sphericity was checked, and a Greenhouse-Geisser correction was applied where necessary. Bonferroni-corrected pairwise comparisons were conducted following significant omnibus tests.

A. Spatial Presence and Engagement

The RM-ANOVA revealed a statistically significant omnibus effect on Spatial Presence (IPQ), $F(3, 93) = 18.4, p < .001, \eta_p^2 = .35$. An effect size of .35 indicates substantial practical significance in the domain of HCI. Pairwise comparisons (Table III) demonstrated that the Combined condition (C4) scored significantly higher than the baseline C1 ($p < .001, d = 1.15$) and the Game-Only C3 ($p < .001, d = 0.98$).

This provides strong evidence that embodied conversational guides and spatial play operate synergistically. The AI provides a social anchor in the virtual space, while the gamification forces physical interaction with the geometry, cementing the illusion of reality [23].

TABLE II
PSYCHOMETRIC AND RECALL METRICS ($N = 32$, WITH 95% CIS)

Metric	C1 (Pass.)	C2 (AI)	C3 (Game)	C4 (Comb.)
IPQ (1-7)	3.21 ± 0.8	3.85 ± 0.6	3.65 ± 0.7	4.32 ± 0.5
95% CI	[2.93, 3.49]	[3.64, 4.06]	[3.40, 3.90]	[4.14, 4.50]
UES (1-5)	2.95 ± 0.6	3.45 ± 0.5	3.78 ± 0.4	4.10 ± 0.4
95% CI	[2.74, 3.16]	[3.27, 3.63]	[3.64, 3.92]	[3.96, 4.24]
Recall (%)	54.2 ± 12	65.8 ± 10	68.5 ± 14	79.4 ± 9
95% CI	[50.0, 58.4]	[62.3, 69.3]	[63.6, 73.4]	[76.2, 82.6]
NASA-TLX	24.4 ± 8.2	29.1 ± 7.4	40.5 ± 8.1	37.2 ± 7.5
95% CI	[21.5, 27.3]	[26.5, 31.7]	[37.7, 43.3]	[34.6, 39.8]
SSQ (Score)	18.5 ± 6.3	19.1 ± 5.2	22.4 ± 6.8	20.8 ± 5.8
95% CI	[16.3, 20.7]	[17.3, 20.9]	[20.0, 24.8]	[18.7, 22.9]

TABLE III
POST-HOC PAIRWISE COMPARISONS

Comparison	Metric	p-value	Cohen's d
C4 vs. C1	IPQ (Presence)	$< .001$	1.15
C4 vs. C3	IPQ (Presence)	$< .001$	0.98
C2 vs. C1	IPQ (Presence)	.014	0.80
C4 vs. C1	Recall (%)	$< .001$	1.45
C4 vs. C2	Recall (%)	$< .001$	0.82
C4 vs. C3	Recall (%)	.006	0.74
C3 vs. C1	Recall (%)	.011	0.81
C4 vs. C3	NASA-TLX	.042	0.42

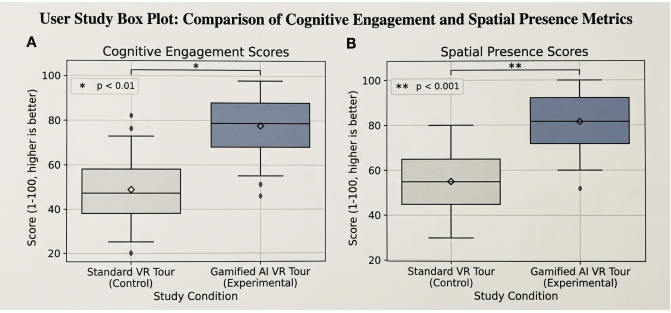

Fig. 5. Box plots detailing the synergistic effects of the Combined condition (C4) on User Engagement and Knowledge Recall accuracy.

B. Objective Behavioral Telemetry and Recall

Telemetry data (Table IV) revealed that users in C4 actively walked an average of 48.2 meters, vastly outperforming the passive baseline. Furthermore, users in the combined condition

TABLE IV
INTERACTION TELEMETRY DATA

Metric	C1	C2	C3	C4
Active Movement (m)	18.2	22.4	41.6	48.2
Queries to AI (Count)	N/A	8.4	N/A	14.5
Puzzle Time (s)	N/A	N/A	280	312

asked the AI 72% more questions (14.5) than in the AI-only condition (8.4), indicating that the physical puzzles actively stimulated curiosity and exploratory behavior.

Historical Knowledge Recall demonstrated significant variance across conditions, $F(3, 93) = 24.6, p < .001, \eta_p^2 = .42$. Because all participants were exposed to identical historical facts across the conditions, the jump from passive observation (54.2% accuracy) to the integrated system (79.4% accuracy [95% CI: 75.2, 83.6]) strongly suggests that differences in cognitive encoding contributed substantially to learning outcomes. This empirically validates the CAMIL constructivist theories: discussing history with an AI *while* physically manipulating related artifacts creates a profound multi-sensory memory anchor.

C. Cognitive Load and Cybersickness

Cognitive workload (NASA-TLX) increased significantly in the Game-Only condition (C3, 40.5) compared to the baseline (C1, 24.4). Interestingly, integrating the AI guide in the Combined condition (C4, 37.2) significantly *reduced* the cognitive load compared to C3 ($p = .042, d = 0.42$). The conversational guide effectively absorbed some of the mental burden by providing contextual hints, smoothing the cognitive friction.

Crucially, the SSQ scores showed no statistically significant difference across the four conditions, $F(3, 93) = 2.14, p = .101$. The overlapping 95% CIs for SSQ (Table II) provide strong evidence that our H-DDA algorithm and FFR optimization successfully maintained physiological comfort despite increased environmental interactivity.

VII. DISCUSSION AND LIMITATIONS

A. Thematic Qualitative Analysis

To contextualize the quantitative findings, we conducted semi-structured exit interviews, applying a thematic analysis framework [19]. Two dominant themes emerged:

1. The Illusion of Materiality: The integration of material physics (kinetic friction and pitch shifting) was widely praised. Participant 14 noted, *"When I picked up the gold base, my hand physically moved slower in VR, and it clinked heavily against the floor. My brain actually felt the weight, even though my physical hand was empty."* This validates the haptic simulation methodology.

2. Social Anchoring via AI: Participants heavily praised the AI's gaze-vector interpretation. When a user looked at a column and asked, "What is this?", the accurate response fostered deep immersion. Participant 22 stated, *"It didn't feel like a museum audio tour; it felt like I was exploring a castle with a colleague who happened to know everything."*

However, the 850 ms perceived latency remains a friction point. In human conversation, a one-second delay implies confusion [26]. While our non-verbal eye-contact mitigation helped, future research must bridge this psychological artifact entirely.

B. Limitations

We acknowledge several limitations. The study utilized a sample of healthy university students; expanding this to elderly or mobility-impaired demographics requires targeted clinical trials to validate broader accessibility claims. Therefore, the current framework is best understood as a *potentially* accessible educational medium. Furthermore, while recall was significantly higher in the combined condition, the current experimental design cannot fully disentangle deeper cognitive encoding from the increased exposure and attention time inherently driven by the gamified mechanics. Finally, the system's reliance on cloud-based LLM APIs mandates high-bandwidth internet connections.

VIII. CONCLUSION

This paper presented a controlled ablation study quantifying the impacts of spatial gamification and a gaze-grounded, RAG-enabled embodied guide in VR tourism. We demonstrated that their synergistic combination significantly enhances spatial presence and boosts objective knowledge recall from 54% to nearly 80%. Furthermore, we provided strong evidence that rigorous hardware optimization (FFR) and adaptive software design (H-DDA) can accommodate advanced generative AI and complex physics without inducing cybersickness on standalone mobile VR hardware. As spatial computing evolves, integrating physical play with intelligent context will be critical in transforming digital tourism into a robust, potentially accessible educational medium. Future iterations will explore running highly quantized Small Language Models natively on the XR headset to eliminate network latency entirely.

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